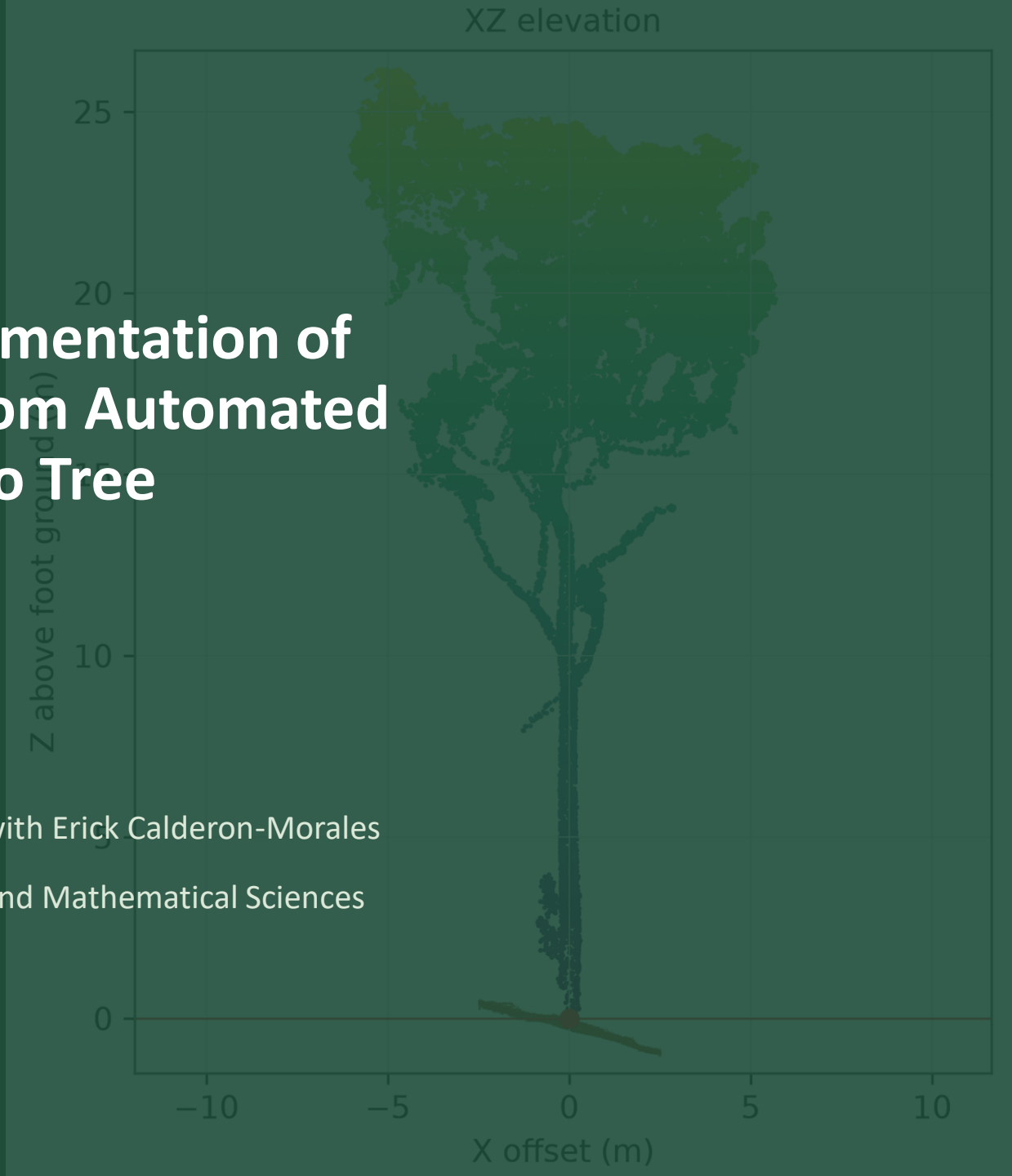
WESTERN SYDNEY UNIVERSITY

Semi-Automated Individual-Tree Segmentation of Blue Mountains TLS Point Clouds: From Automated Extraction and Manual Refinement to Tree Measurements



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INFO6003 Postgraduate Research Project · School of Computer, Data and Mathematical Sciences
Western Sydney University · Spring 2026

Tree measurements need every point assigned to the right tree — and that is hard to automate



TLS = terrestrial laser scanning. DBH = stem diameter at 1.3 m.

Merged crown
one tree inflated, one tree lost

TLS captures stems, branches and crowns.

Segmentation is the hardest step to automate where eucalypt crowns overlap.

No published benchmark covers temperate eucalypt forest.

Four questions follow the workflow from segmentation to field validation

RQ1

What does automatic segmentation get wrong, and what does correction change?

RQ2

How much manual effort does correction need?

RQ3

How many corrected trees meet quality criteria for height, DBH and crown — and why do the others fail?

RQ4

How does TLS DBH compare with the field inventory?

Contributions

C1 163-tree corrected dataset

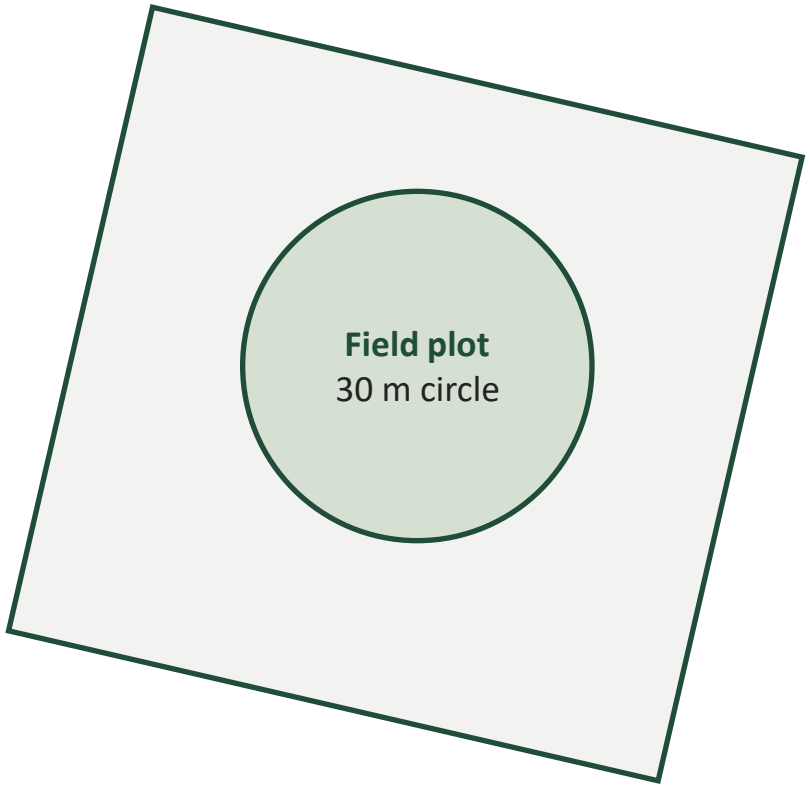
C2 Quality-labelled measurements for every tree

C3 DBH comparison with the located field plot and tree-by-tree matching

Existing tools used unchanged. Biomass deferred (see conclusion).

DBH = stem diameter at 1.3 m

A 0.32 ha TLS crop that contains a 30 m field inventory plot



Schematic: crop rotated about 13°, field plot inside

TLS

Plot WSF_ML2A · scanned 12 May 2025 at 1 cm

Crop 57.3 × 55.0 m

234 M points → thinned to 3 cm: 45.4 M

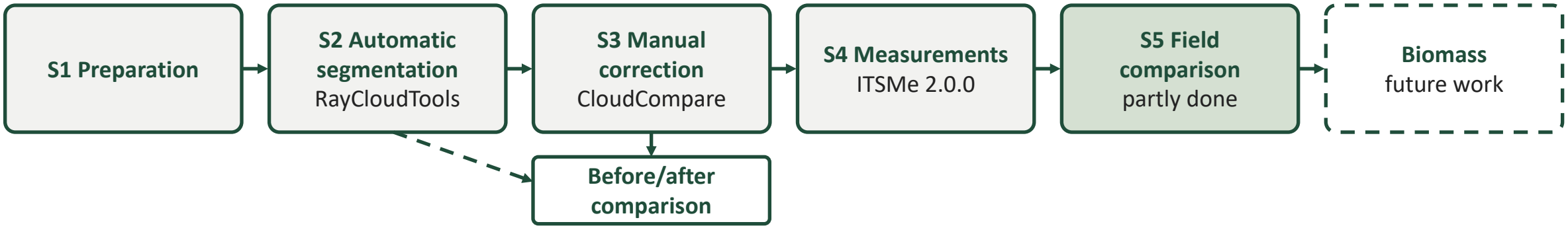
Field inventory

30 m circle (0.071 ha) · surveyed 3 Apr 2024, 13 months earlier

26 live trees, DBH ≥ 11.2 cm

No tree coordinates, no heights · approximate site map

A five-stage workflow — each decision chosen for a reason



Decision	Why
Thin to 3 cm	Manageable processing; trade-off: loses fine branches and thin stems
Rayextract + manual correction	Best benchmark F1, but automation alone is insufficient in overlapping crowns
VALID / REVIEW / N/A labels	Make quality visible instead of silently mixing good and doubtful values
Estimated DBH kept separate	Never mixed with measured DBH
IoU ≥ 0.5 before/after protocol	Same protocol as the benchmark (Cherlet et al., 2026)
Position first, DBH second for matching	Species not visible in scan (supervisor advice)

RQ1

162 → 163 trees, but correction changed the segmentation substantially

162 automatic trees → **86** identical | **29** automatic trees split | **25** corrected trees merged from several → **163** corrected trees

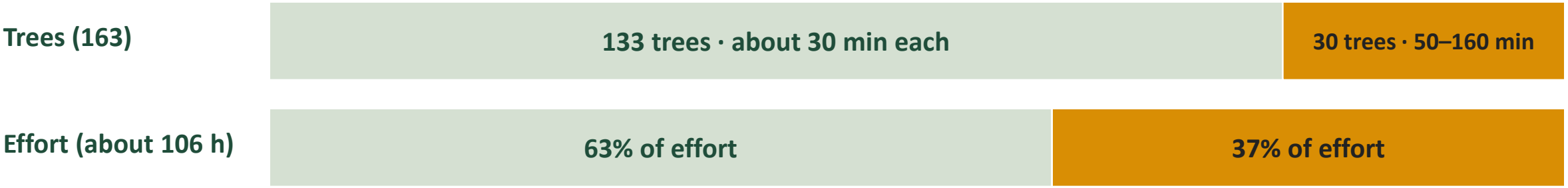
0.85 F1 at IoU ≥ 0.5 (0.62 at 0.9)	**8.5%** of points removed	**12.6%** of points moved to another tree	**36.8%** of points left unassigned by automation

IoU = overlap of shared points. F1 = combined precision and recall.

Evidence: Measures change, not accuracy — the reference was made by editing the automatic output.

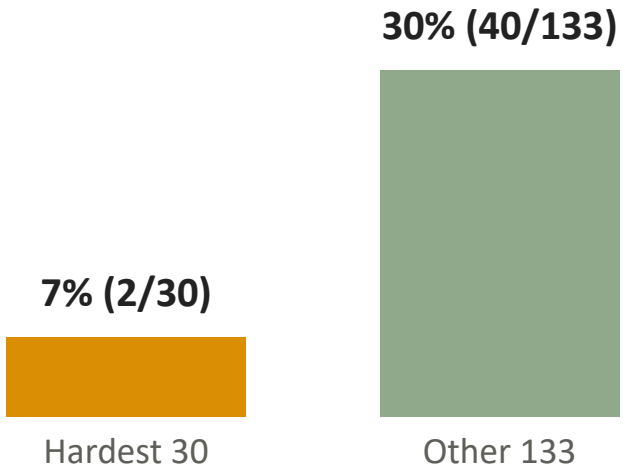
RQ2

About 106 hours — over a third spent on 18% of trees



30 trees (18%) → 37% of effort

DBH VALID rate

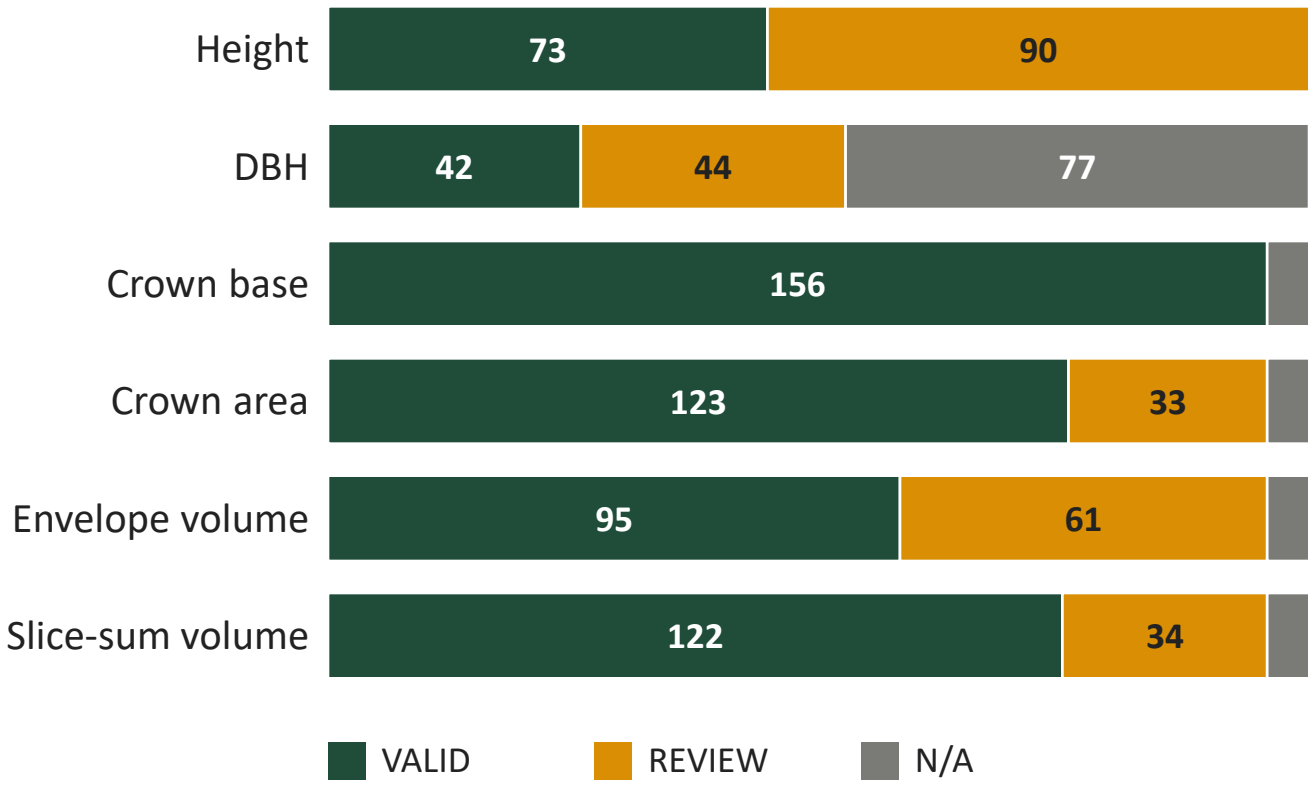


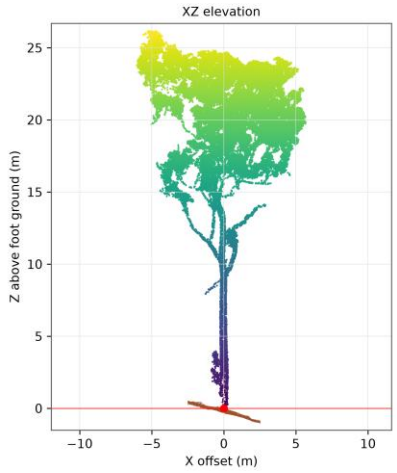
Fisher's exact test
p = 0.01

Evidence: effort is a per-tree estimate made afterwards, not a timed log; acceptance was by visual judgement.

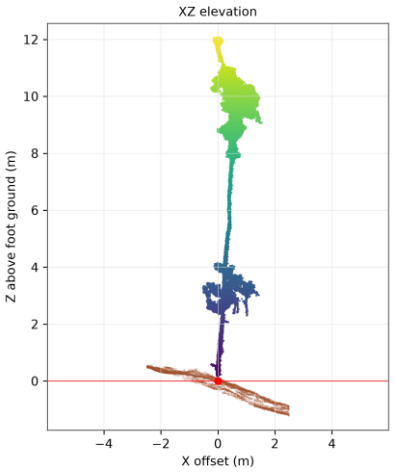
RQ3

Only 16 of 163 trees meet every quality criterion





tree_069 · 26.2 m
DBH 51.7 cm VALID
Crown area 97.6 m²
REVIEW



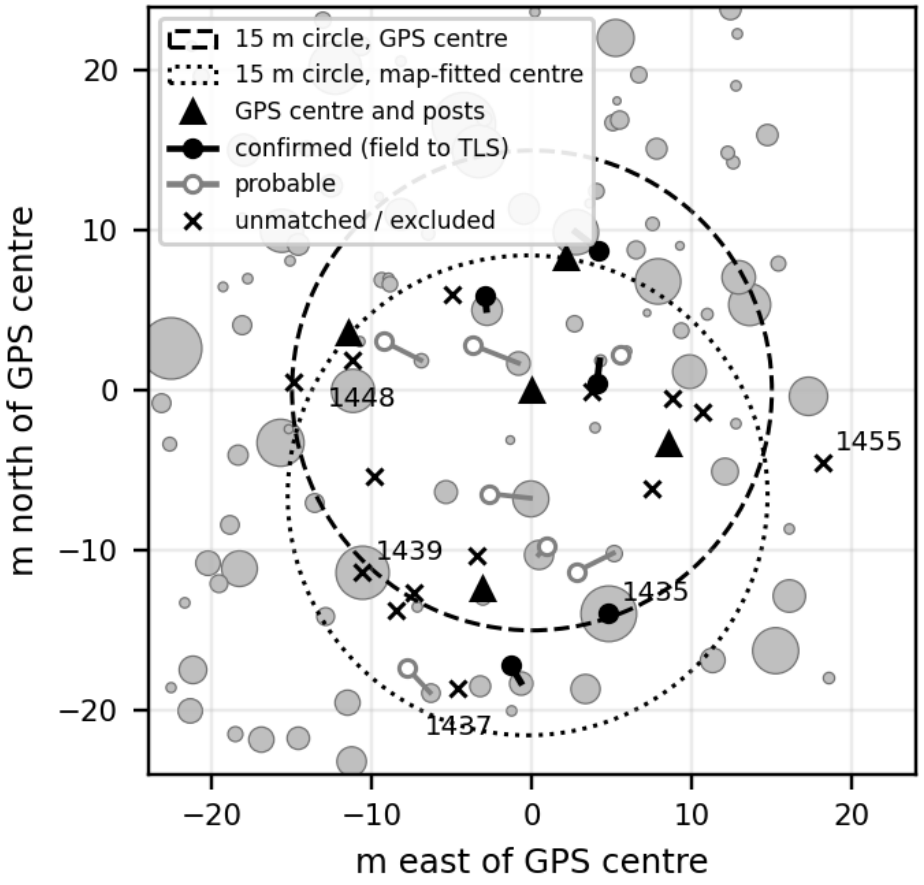
tree_146 · 12.1 m
DBH 10.7 cm VALID
Crown area 1.9 m²
REVIEW

16 **trees with height, DBH, crown area and crown volume all VALID**

Evidence: VALID trees are not a random sample — their medians are not plot averages.

RQ4

The field plot lies inside the crop, and DBH distributions are indistinguishable



Measured TLS DBH vs field median 32.5 cm

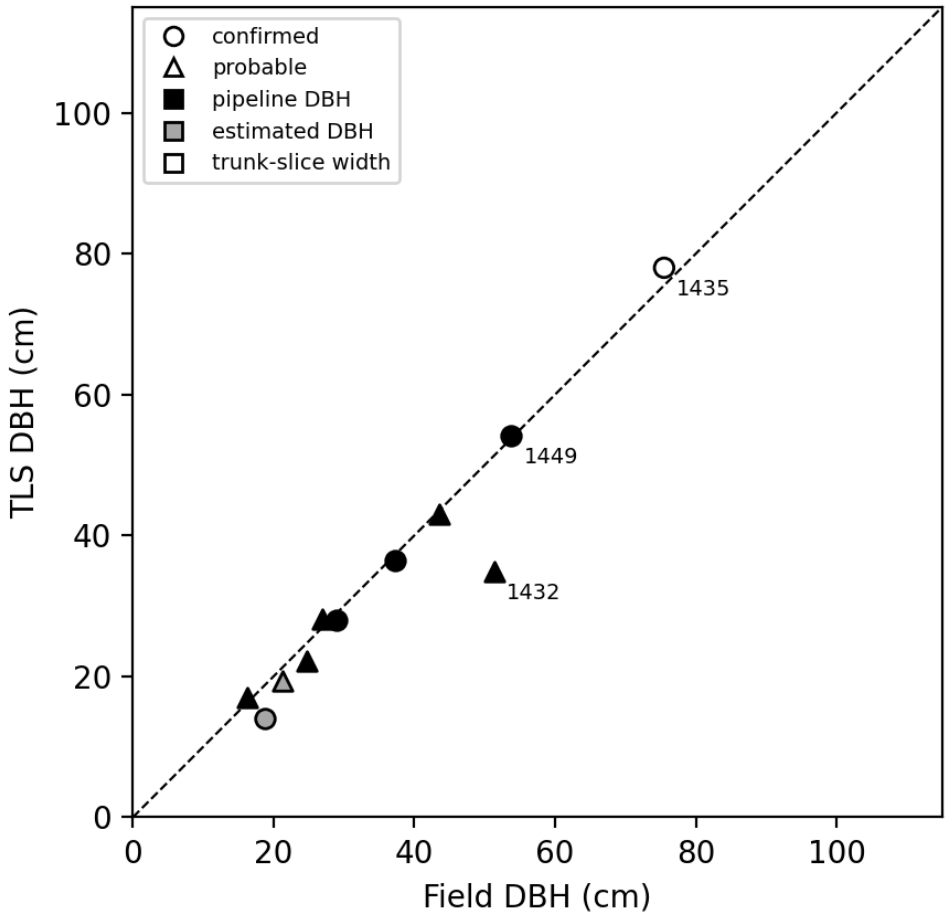
Plot centre	n	TLS median	Difference	p
GPS centre	28	30.0 cm	−7.7%	0.62
Map-fitted centre	20	28.0 cm	−13.8%	0.48

Evidence: no significant difference (p = 0.35–0.91 across all six sets), but low power with 10–33 trees.

Survey posts lie 8.5–12.8 m from the GPS centre instead of 15 m, so GPS is off by several metres. A second, map-fitted centre was tested.

RQ4

5 tagged trees matched with confidence; the largest trees reveal different failure modes



5 confirmed	**8** probable	**12** unmatched
p ≈ 0.007 shuffled-DBH test	**2.4 cm** confirmed RMSE	**1** excluded

Largest field trees

1435 (75.5 cm): oval trunk, circle fit too low → measurement

1448 (83 cm): two intertwined trunks → segmentation

1439 (90 cm): measured 63.6 cm → unexplained

1437 (107.5 cm), 1455 (74.5 cm) → not identified

Evidence: matches required DBH within 25%, so the RMSE is optimistic. One confirmed tree is an anchor; 4 are independent.

Segmentation success does not guarantee usable measurements

Measurement accuracy
DBH only; small, partly selected sample

Measurement success
measured: the main finding

Segmentation change
measured; no independent accuracy reference

Evidence ladder: what each result can and cannot show

Limitations

- Visual acceptance by two people
- GPS errors of several metres, approximate map, two anchors
- 13-month gap, survey to scan
- Effort estimated afterwards
- Some settings unrecorded
- One 0.32 ha crop

Next: confirm matches in the field, then biomass

Done

- 163 corrected trees
- Quality-labelled measurements for all
- Field plot located
- 5 confirmed matches
- Reproducible code

Next

- Field-check matches; locate 1437 and 1455 (12-tree sheet ready)
- Per-tree accept/reject records
- Refit DBH at full resolution (234 M points)
- Then biomass

Take-home

Tree counts and segmentation success can both hide problems — quality labels make them visible.

Use of AI: generative AI assistants (OpenAI Codex, Anthropic Claude) supported script development, report editing and preparation of these slides, declared per WSU Academic Integrity Policy. Manual segmentation, anchor checks and all analytical decisions are my own. Code: github.com/Thinhrichkidz/tls-tree-segmentation (commit 72479a1)

Thank you — questions?

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